

ADAM TAY XUE HAO

AI Algorithm Engineer

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PROFESSIONAL SUMMARY

AI Engineer with ~3 years of experience developing and optimizing **machine learning and large language models** for real-world applications. Experienced in LLM fine-tuning, quantization, knowledge distillation, RAG, Text-to-SQL, and model evaluation, with hands-on expertise in adapting open-source models for enterprise and edge AI use cases. Proven track record of improving model accuracy and efficiency through data-driven experimentation, training optimization, and performance analysis. Published first author in a **Scopus Q2 journal on Deep Reinforcement Learning** and contributor to award-winning AI solutions showcased at CES 2025 and recognized with the Best AI Debut award at MWC 2025.

WORKING EXPERIENCE

❖ **AI Algorithm Engineer, Mediatek** Singapore | Aug 2025 — Present

Gemma 4 Enablement on Mediatek NPU in collaboration with Google:

- Led the end-to-end porting of Gemma 4 LLM, Vision Encoder, and MTP Speculative Decoding Drafter to Google AICore runtime through a 4-stage lowering pipeline: **Huggingface graph -> NPU-aware Pytorch graph -> PTQ-ed and JIT-traced TFLite -> Compiled NPU model**.
- Developed Google's Android runtime for Gemma 4, including **tensor I/O buffer management, Memory footprint optimization, KV Cache management, batch inference, and accuracy debugging analytics**. Passed all of Google's accuracy evaluation and Prompt-mode latency KPI (**~ 160% for the Fast model, ~130% for the Main model**).
- Implemented Matformer based on the research paper to enable weight sharing, optimize disk and memory overhead by ~50%.

Mediatek LLM SDK:

- Contributed to Mediatek's LLM SDK, focusing on converting customer's model graphs into NPU compatible graph and QAT with Deepspeed to recover accuracy loss after quantization.
- Implemented a **Two-phase knowledge distillation style Training** to recover model compilation accuracy loss of LLM models.
- Developed tensor-level and logits-level analytics visualization tool to explain/debug distribution or cosine similarity shifts throughout the 4-stage pipeline or QAT/DAT.

❖ **AI System Engineer, Intel Corporation** Penang | Jan 2024 — Aug 2025

Intel AI Assistant Builder:

- Core AI developer of an **edge AI solution**, building a fully **local LLM** application using **LlamaIndex** in **Python**, developed features such as RAG, Text-to-SQL, Summarization, Resume Scoring, and Agentic workflows exposed through **gRPC** API.
- Extended Huggingface's Transformers library to perform multi-token zero-shot classification, enabling a more efficient and deterministic output; Work submitted for **patent filing**.
- Enabled Multimodal Large Language Models using Optimum intel for image querying and generation.
- Developed and maintained a CICD pipeline using GitHub Action for automated building and testing.
- Project featured in CES 2025 and received the Best AI Debut award at MWC 2025.

NPU Kernel Optimization with Boolean Packing:

- Optimized **memory footprint** of the Select kernel on NPU by **representing Booleans in INT1** using C++,

achieving **153% performance boost** in ResNet50 performance on Intel NPU.

LLM Code Generation Fine-tuning:

- Performed PEFT (Parameter Efficient Fine-tuning) using **LoRA** with **Unsloth** to fine-tune domain specific code (Device Modelling Language) generation, achieving **84.8%** accuracy after finetuning using **LLM-as-a-judge** + **Domain Expert Evaluation**. Accuracy of the **base model** was **36.1%**.

Initiatives / Proof of Concepts:

- Led 2 junior engineers to migrate from naïve chunk size-based chunking to HTML aware chunking to reduce chunk fragmentation, leading to **37.5%** overall improvements in accuracy using **LLM-as-a-judge** with GPT-4o.
- Initiated a learning group consisting of 3 junior engineers to study model graph partitioning techniques using the Openvino NPUW framework to enable day 0 LLM support on Intel NPU.

❖ **Freelance AI Engineer, Fiverr** Remote | Aug 2023 — Dec 2023

Secured 3 repeat orders from a client:

- Conducted PEFT using QLoRA to fine-tune Falcon 7b on client's mental health dataset.
- Performed data preparation for fine-tuning by converting raw text into JSON format with prompt-response pairs.
- Applied semi-supervised learning with self-training for MBTI personality prediction on a 20:80 labeled-to-unlabeled dataset of Twitter comments. Achieved **0.91 average f1-score across 4 categories**.

EDUCATION

❖ **Tunku Abdul Rahman University of Management and Technology** May 2021 — Aug 2023

Bachelor of Computer Science (Honours), Data Science

Kuala Lumpur

- Grade: **4.0/4.0 CGPA**
- Top 1 Graduate | Full scholarship | Book Prize Award Winner | First Class Honours

PUBLICATIONS

❖ **Publication**

- Main author. Paper accepted by a Scopus Q2 Journal.
- Tay, X., H. & Lim, S., M. (2024, Sep). **Deep Reinforcement Learning in Cryptocurrency Trading: A Profitable Approach**, Journal of Telecommunications and the Digital Economy, Vol. 12 No. 3, 126-147

TECHNICAL SKILLS

❖ **Core Skills**

AI & ML: Transformer Architectures, LLM, RAG, Fine-tuning, Model Optimization and Enablement, Data Science, Machine Learning, Reinforcement Learning, Data Visualization and Analytics

Software Engineering: Backend development, System design, Stateless API design, SOLID/DRY

❖ **Programming Languages**

Python, C++, SQL

❖ **Tools and Frameworks**

Pytorch, Git, GitHub Actions, Docker, Bash scripting, gRPC, REST, AWS/Azure, React.js

❖ **Natural Language**.....

English, Mandarin, Malay